

Final Project Initial Analysis

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2024-02-29

Dataset: PEW RESEARCH CENTER'S AMERICAN TRENDS PANEL WAVE 105 socialDemo

Load data:

```
socialDemo <- read_csv("socialDemo.csv")
```

```
## Rows: 3581 Columns: 159
## -- Column specification -----
## Delimiter: ","
## chr (153): DEVICE_TYPE_W105, LANG_W105, FORM_W105, XTABLET_W105, ECON_SIT_W1...
## dbl (6): QKEY, INTERVIEW_START_W105, INTERVIEW_END_W105, FPKNOW_W105, WEIG...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Variables

Research Question: What effect has social media had on Democracy in the eyes of American citizens? Is American satisfaction with democracy in the United States related to their opinions on social media's impact on democracy?

Hypothesis: Low American satisfaction with democracy in the US is correlated with belief that social media has a negative impact on democracy. (and vice versa)

Explanatory/Independent variable: Individual Americans' satisfaction with democracy in the US

Outcome/Dependent variable: Individual Americans' opinions on social media's impact on democracy

Variables of interest:

SATISFIED_DEMOCRACY_w105: How satisfied are you with the way democracy is working in the U.S?
1) Very satisfied 2) Somewhat satisfied 3) Not too satisfied 4) Not at all satisfied

F_PARTY_FINAL: As of today do you lean more to 1) Republican 2) Democrat 3) Independent 4) Something else

SNSUSE: Do you use social media sites like Facebook, Twitter, or Instagram? 1) Yes, I use social media sites 2) No, I do not use social media sites

SOCIAL_DEMOCRACY_w150: Overall, when you consider all the advantages and disadvantages of social media, would you say social media has been 1) More of a GOOD THING for democracy in the U.S. 2) More of a BAD THING for democracy in the U.S.

SMIMPACT_BATT: In general, do you think access to the internet and social media has made people more or less SMIMPACT_INFORMED_w105: Informed about current events in our country? SMIMPACT_DIVIDED_w105: Divided in their political opinions? SMIMPACT_MANIPULATE_w105: Easy

to manipulate with false information and rumors? SMIMPACT_DISCOURSE_w105: Civil in the way they talk about politics? 1) More 2) Less 3) Not had much impact

New columns for making responses numeric: SATISFIED_DEMOCRACY

```
socialDemo$SATISFIED_DEMOCRACY <- (ifelse(socialDemo$SATISFIED_DEMOCRACY_W105 == "Very satisfied", 4,
  ifelse(socialDemo$SATISFIED_DEMOCRACY_W105 == "Somewhat satisfied",
  ifelse(socialDemo$SATISFIED_DEMOCRACY_W105 == "Not too satisfied",
  ifelse(socialDemo$SATISFIED_DEMOCRACY_W105 == "Not at all satis
```

PARTY

```
socialDemo$PARTY <- (ifelse(socialDemo$F_PARTY_FINAL == "Republican", 1,
  ifelse(socialDemo$F_PARTY_FINAL == "Democrat", 2,
  ifelse(socialDemo$F_PARTY_FINAL == "Independent", 3,
  ifelse(socialDemo$F_PARTY_FINAL == "Something else", 4, NA))))
```

SOCIALMEDIA_USE

```
socialDemo$SOCIALMEDIA_USE <- (ifelse(socialDemo$SNSUSE_W105 == "Yes, I use social media sites", 1,
  ifelse(socialDemo$SNSUSE_W105 == "No, I do not use social media sites",
```

DEMOCRACY

```
socialDemo$DEMOCRACY <- (ifelse(socialDemo$SOCIAL_DEMOCRACY_W105== "More of a GOOD THING for democracy",
  ifelse(socialDemo$SOCIAL_DEMOCRACY_W105 == "More of a BAD THING for democracy
```

INFORMED Effect of social media on education of current events

```
socialDemo$INFORMED <- (ifelse(socialDemo$SMIMPACT_INFORMED_W105 == "More", 1,
  ifelse(socialDemo$SMIMPACT_INFORMED_W105 == "Less", 2,
  ifelse(socialDemo$SMIMPACT_INFORMED_W105 == "Not had much impact", 3, NA)))
```

DIVISION Effect of social media on political division

```
socialDemo$DIVISION <- (ifelse(socialDemo$SMIMPACT_INFORMED_W105 == "More", 1,
  ifelse(socialDemo$SMIMPACT_INFORMED_W105 == "Less", 2,
  ifelse(socialDemo$SMIMPACT_INFORMED_W105 == "Not had much impact", 3, NA)))
```

MANIPULATE Effect of social media on manipulation through false information

```
socialDemo$MANIPULATE <- (ifelse(socialDemo$SMIMPACT_MANIPULATE_W105 == "More", 1,
  ifelse(socialDemo$SMIMPACT_MANIPULATE_W105 == "Less", 2,
  ifelse(socialDemo$SMIMPACT_MANIPULATE_W105 == "Not had much impact", 3, NA)))
```

DISCOURSE Effect of social media on civil political discourse

```
socialDemo$DISCOURSE <- (ifelse(socialDemo$SMIMPACT_DISCOURSE_W105 == "More", 1,
  ifelse(socialDemo$SMIMPACT_DISCOURSE_W105 == "Less", 2,
  ifelse(socialDemo$SMIMPACT_DISCOURSE_W105 == "Not had much impact", 3, NA)))
```

Summary of our numerical variables

Name	Description
'SATISFIED_DEMOCRACY	' How satisfied respondent is with US democracy, coded on a scale of 1 to 4, with 4 being “
PARTY	The political party the respondent identifies with, coded ...
SOCIALMEDIA_USE	Binary variable coded “1” if respondent uses social media and “0” if not
INFORMED	The...
DIVISION	The...
MANIPULATE	The...
DISCOURSE	The...

Summary for American’s view on social media’s effect on democracy

```
summary(socialDemo$DEMOCRACY)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
## 0.0000  0.0000  0.0000  0.3344  1.0000  1.0000    64
```

Mean of 0.3344 means a vast majority of respondents think social media has been a bad thing for democracy. There are 64 missing values.

```
summary(socialDemo$SATISFIED_DEMOCRACY)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
## 1.000  2.000  2.000  2.229  3.000  4.000    20
```

Mean of 2.229 means most respondents are on average “not too satisfied” with the US democracy. There are 20 respondents who did not respond to this question.

Visualization or Regression

```
mod1 <- lm(SATISFIED_DEMOCRACY ~ DEMOCRACY, data = socialDemo)
coef(mod1)
```

```
## (Intercept)  DEMOCRACY
## 2.1127787    0.3402929
```